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 Roll No: MA25BTECH11007

Date: 03.09.2025

Time: 90 minutes

16. What is the reason for ozone layer depletion by chlorofluorocarbons (CFCs)  
 a) C-F bond cleavage by UV light ☒ b) C-Cl bond cleavage by UV light ☒ c) Adsorption of  $O_3$   
 d) complete dissociation
17. Which statement is correct in the solvent-based direct air-capturing method  
 a) KOH is regenerated by the reaction of  $K_2CO_3$  with  $Ca(OH)_2$   
 b)  $Ca(OH)_2$  is regenerated by the reaction of  $CaO$  with  $H_2O$ , which is essential for DAC method  
 c) neither a nor b ☒ d) Both a and b
18. What is the fraction of atmospheric window  
 a) 15-30% ☒ b) 45-60% ☒ c) 70-75% ☐ d) 70-85% ☐
19. Bioethanol is produced via fermentation of  
 a) Lignocellulose ☒ b) Starch ☒ c) Lipids ☐ d) All of these ☐
20. Photocatalysts convert  $CO_2$  to valuable chemical such as methanol via  
 a) electron transfer ☒ b) proton transfer ☐  
 c) electron transfer coupled proton transfer ☒ d) Hydride transfer ☐
21. Biodiesel is nothing but  
 a) Vegetable oil ☒ b) alkyl esters of fatty acids ☒ c) Triglycerides ☐ d) Hydrocarbons ☐
22. The primary component of the plant cell wall is  
 a) Starch ☒ b) Cellulose ☒ c) hemicellulose ☐ d) lignin ☐
23. The intensity of the solar radiation received by the Earth from the Sun is  
 a) 384 yW ☒ b) 1300 kW ☐ c) 1400 W ☒ d) 1300 kW ☐ (C)
24. If the wind velocity in a given area increases from 10 km/hour to 20 km/hour, how much wind power increases by?  
 a) 2-fold ☒ b) 4-fold ☒ c) 8-fold ☒ d) No change ☐
25. World Environmental day is  
 a) 5<sup>th</sup> July ☒ b) 5<sup>th</sup> June ☒ c) 15<sup>th</sup> July ☐ d) 15<sup>th</sup> June ☐
26. What is the active metal center of particulate methane monooxygenase (p-MMO) enzyme?  
 a) Fe ☒ b) Cu ☒ c) Co ☐ d) Mo ☐
27. Which of the following compounds does not belong to BTEX?  
 a) Benzene ☒ b) Toluene ☒ c) Naphthalene ☒ d) Ethyl benzene ☐
28. Where does beta-oxidation primarily occur in eukaryotic cells?  
 a) Cytoplasm ☒ b) Nucleus ☐ c) Mitochondrial matrix ☒ d) Golgi apparatus ☐
29. How many acetyl-CoA molecules are produced from complete beta-oxidation of palmitic acid ( $C_{16}$ )?  
 a) 6 ☒ b) 8 ☒ c) 10 ☐ d) 9 ☐

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43. In a Lineweaver-Burk plot, competitive inhibition results in:
- ☒ a) No change in y-intercept
  - ☐ b) Change in y-intercept
  - ☐ c) No change in slope
  - ☐ d) Parallel lines
44. The Lineweaver-Burk plot is a graphical representation of enzyme kinetics that helps to:
- ☐ a) Measure ATP production
  - ☐ b) Identify the type of inhibition
  - ☐ c) Monitor protein folding
  - ☒ d) Determine the amino acid sequence
45. The principle "Prevention" in green chemistry means:
- ☐ a) Preventing temperature fluctuations
  - ☒ b) Preventing waste formation rather than treating it later
  - ☐ c) Preventing product formation
  - ☐ d) Preventing the use of glassware
46. An example of a green catalyst is:
- ☐ a) Mercury
  - ☐ b) Lead
  - ☒ c) Enzyme
  - ☐ d) Cyanide
47. Consider the reaction:  $\text{CH}_3\text{CH}_2\text{Br} + \text{NH}_3 \rightarrow \text{CH}_3\text{CH}_2\text{NH}_2 + \text{HBr}$   
 Calculate the atom economy for producing ethylamine ( $\text{C}_2\text{H}_5\text{NH}_2$ ). (Molar masses:  $\text{CH}_3\text{CH}_2\text{Br} = 108.97$ ,  $\text{NH}_3 = 17.03$ ,  $\text{CH}_3\text{CH}_2\text{NH}_2 = 45.08$ )
- ☐ a) 100%
  - ☐ b) 41.4%
  - ☒ c) 36.6%
  - ☐ d) 55.5%
48. What is the primary goal of industrial ecology?
- ☐ a) To improve profit margins for industries
  - ☐ b) To eliminate all industrial activity
  - ☒ c) To optimize resource use and minimize environmental impact
  - ☐ d) To produce more goods with higher energy use
49. Industrial ecology views industrial systems as analogous to:
- ☐ a) Political systems
  - ☒ b) Ecosystems
  - ☐ c) Economic markets
  - ☐ d) Thermodynamic systems
50. The Kalundborg Industrial Symbiosis project is an example of:
- ☐ a) Failed green innovation
  - ☐ b) Open-loop manufacturing
  - ☒ c) Practical application of industrial ecology
  - ☐ d) Renewable energy technology



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45  
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Please tick the correct answer.

1. Which one is the highest abundant element in the Earth crust  
 a) Iron ☒ b) Oxygen ☒ c) Silicon ☐ d) Magnesium ☒ (b)
2. If the chloride ion (molar mass is 35.45 g/mol) concentration (molarity) in the ocean water is 560 mM, it corresponds to the ppm scale of  
 a) 19900 ☒ b) 1990 ☐ c) 199 ☐ d) 199000 ☒ (a)
3. The order of particulate matter (carbon),  $N_2O$ ,  $CH_4$ ,  $SF_6$  in terms of residence time in the atmosphere is  
 a)  $C < N_2O < CH_4 < SF_6$  ☒ b)  $C < CH_4 < N_2O < SF_6$  ☒ c)  $N_2O < C < CH_4 < SF_6$  ☐ d)  $CH_4 < N_2O < C < SF_6$  ☐ (b)
4. In steam methane reforming process, how many grams of hydrogen get produced by reacting 32 grams of  $CH_4$   
 a) 4 g ☒ b) 12 g ☒ c) 8 g ☐ d) 2 g ☐ (b)
5. If global annual energy consumption is 500 exa Joules, it corresponds to power consumption in Watts  
 a)  $15.85 \times 10^{12}$  ☒ b)  $1585 \times 10^{12}$  ☐ c)  $15.85 \times 10^9$  ☐ d)  $1585 \times 10^9$  ☐ (a)
6. What is the ionic charge carrier that shuttles from anode to cathode in direct methanol fuel cell  
 a) methoxide ion ☒ b) methyl cation ☐ c) hydroxyl ion ☒ d) proton ☒ (d)
7. Splitting of water to produce hydrogen is a  
 a) favored thermodynamically ☒ b) unfavored thermodynamically ☒ c) unfavored kinetically ☐ d) unfavored thermodynamically and kinetically both ☐ (b)
8. Solar thermal plants use the following for light energy conversion  
 a) p-n junction diode ☒ b) semiconductor ☒ c) parabolic dish ☐ d) flat plate glass ☒ (a)
9. What is the intermediate nucleus in nuclear fission of  $^{235}U$ ?  
 a)  $^{236}U$  ☒ b)  $^{238}U$  ☐ c)  $^{144}Ba$  ☐ d)  $^{89}Kr$  ☒ (a)
10. The major composition of liquid petroleum gas includes  
 a) Propene and butane ☒ b) Propane and butane ☒ c) Propane and butene ☐ d) Propene and butene ☐ (b)
11. Biodiesel can be produced from  
 a) Vegetable oil ☒ b) Animal fat ☒ c) Algae ☒ d) All of these ☒ (d)
12. What is the catalyst used in production of biodiesel?  
 a) Iron ☒ b) Nickel ☒ c) Sodium hydroxide ☒ d) Yeast ☐ (c)
13. Production of Urea from ammonia and water involves  
 a) Hydrolysis ☒ b) dehydration ☒ c) redox reaction ☐ d) hydrogenation ☐ (b)
14. What is the technique used for determination of ocean  $CO_2$  concentration  
 a) Potentiometry ☒ b) Diffraction ☐ c) microscopy analysis ☐ d) amperometry ☐ (a)
15. When the size of the carbon particles (assuming spherical shape) emitted from Diesel engines get reduced by ten times of their size, what would be the increase in the surface area to volume ratio  
 a) 100 ☒ b) 1000 ☐ c) 10 ☐ d) 10000 ☐ (a)

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30. What happens to the propionyl-CoA formed during the oxidation of odd-chain fatty acids?
- It is excreted in urine
  - It is converted to glucose via pyruvate
  - It is converted to succinyl-CoA and enters the TCA cycle
  - It is converted back to fatty acids
31. What is the first molecule that combines with Acetyl-CoA to produce citrate in the Krebs cycle?
- Fumarate
  - Citrate
  - Oxaloacetate
  - Malate
32. What is the monomeric unit in protein?
- Amino acid
  - Lactic acid
  - Glucose
  - Lactate
33. The term "Dirty Dozen" refers to
- 12 banned pharmaceutical drugs
  - 12 persistent organic pollutants
  - 12 greenhouse gases
  - 12 industrial enzymes
34. Which international agreement aims to eliminate or restrict the Dirty Dozen chemicals?
- Kyoto Protocol
  - Montreal Protocol
  - Basel Convention
  - Stockholm Convention
35. What is a major characteristic of the Dirty Dozen pollutants?
- They decompose rapidly in nature
  - They are biodegradable detergents
  - They persist in the environment and bioaccumulate
  - They are used as fertilizers
36. Which of the following is not one of the original Dirty Dozen POPs?
- DDT
  - PCBs
  - Dioxins
  - Carbon dioxide
37. The Arrhenius equation may fail to describe protein reaction rates accurately at
- Moderate temperatures
  - Enzyme optimum pH
  - Extremely high temperatures
  - Low protein concentrations
38. What are xenobiotics?
- Endogenous hormones
  - Nutrients required for growth
  - Foreign compounds not naturally produced by the body
  - Normal metabolic intermediates
39. Which organ is primarily responsible for the metabolism of xenobiotics?
- Kidney
  - Lung
  - Liver
  - Intestine
40. Primary reactions of xenobiotic metabolism typically involve:
- Conjugation
  - Hydrolysis, oxidation, and reduction
  - Methylation
  - Glucuronidation
41. What is the main goal of xenobiotic metabolism?
- Convert fat-soluble substances into energy
  - Increase water solubility for excretion
  - Produce energy from toxins
  - Store them in fat tissues
42. Which of the following best describes competitive inhibition?
- Inhibitor binds to the enzyme-substrate complex only
  - Inhibitor binds irreversibly to the enzyme
  - Inhibitor resembles the substrate and binds to the active site
  - Inhibitor binds to allosteric site of enzyme



| Q# | Answer |
|----|--------|
| 1  | b.     |
| 2  | a      |
| 3  | b      |
| 4  | b.     |
| 5  | a      |
| 6  | d      |
| 7  | d      |
| 8  | c      |
| 9  | a      |
| 10 | b      |
| 11 | d      |
| 12 | c      |
| 13 | b      |
| 14 | a      |
| 15 | c      |
| 16 | b      |
| 17 | d      |
| 18 | a      |
| 19 | b      |
| 20 | c      |
| 21 | b      |
| 22 | b      |
| 23 | c      |
| 24 | c      |
| 25 | b      |

|    |    |
|----|----|
| 26 | b. |
| 27 | c. |
| 28 | c. |
| 29 | b. |
| 30 | c  |
| 31 | c  |
| 32 | a  |
| 33 | b  |
| 34 | d  |
| 35 | c  |
| 36 | d  |
| 37 | c  |
| 38 | c  |
| 39 | c  |
| 40 | b  |
| 41 | b  |
| 42 | c  |
| 43 | a  |
| 44 | b  |
| 45 | b  |
| 46 | c  |
| 47 | c  |
| 48 | c  |
| 49 | b  |
| 50 | c  |